

Spandan Ghoshal

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Research Interests

High-Dimensional Statistics · Change-Point Analysis · Nonparametric Methods · Network Data Analysis · Probabilistic Combinatorics · Statistical Learning Theory

Education

Indian Statistical Institute <i>Ph.D. in Statistics (Senior Research Fellow)</i> Supervisor: Prof. Anil K. Ghosh. Research on high-dimensional change-point detection and nonparametric testing.	Kolkata 2023–Present
Indian Statistical Institute <i>Master of Statistics (M.Stat), 78.9% – First Division with Distinction</i> Specialization: Computational Statistics.	Kolkata 2023
Presidency University <i>B.Sc. in Statistics, CGPA: 8.41/10 – Ranked 1st in Department</i> Batch 2018–2021.	Kolkata 2021
Howrah Zilla School <i>Class XII (WBCHSE), 94.0%</i>	Howrah 2018
Howrah Vivekananda Institution <i>Class X (WBBSE), 88.4%</i>	Howrah 2016

Academic Achievements

- Secured **All India Rank 1** in ISI JRF Entrance Examination (2023).
- Secured **All India Rank 9** in IIT-JAM Statistics (2021).
- Secured **All India Rank 10** in ISI M.Stat Entrance Examination (2021).
- Recipient of the ISI Fellowship for Ph.D. and M.Stat programs.

Publications & Research Work

Manuscripts

Under Review: Ghoshal, S., Ghosh, A. K., and Chakrabarty, A. “High-Dimensional Change-Point Detection Based on Pairwise Distances.” Submitted to *Statistica Sinica*, 2026.

In Preparation: Ghoshal, S. and Ghosh, A. K. “On the large-sample consistency of the distribution-free multivariate two-sample runs test based on the shortest Hamiltonian path.”

Ongoing Projects

Ongoing: Fourth Moment Phenomenon for Edge Statistics in Neighborhood Sampling. Joint work with Prof. Siva Athreya (ICTS) and Prof. Bhaswar B. Bhattacharya (University of Pennsylvania). Initiated at the ICTS-Networks Workshop.

Teaching Experience

Nonparametric and Sequential Methods <i>Teaching Assistant</i> Assisted Prof. Anil K. Ghosh; conducted tutorials and evaluated scripts.	B.Stat 3rd Year, ISI Kolkata Jan–Jun 2026
Statistical Structure in Data <i>Teaching Assistant</i>	PGDBA 1st Year, ISI Kolkata Jul–Dec 2025

Assisted Prof. Anil K. Ghosh for the Post Graduate Diploma in Business Analytics program.

Statistical Data Analysis

ISEC, ISI Kolkata

Co-Instructor

Jan–Jun 2025

Taught international participants in collaboration with Prof. Sumitra Purakayastha.

Descriptive Statistics I

ISEC, ISI Kolkata

Co-Instructor

Jul–Dec 2024

Taught international participants in collaboration with Prof. Partha Sarathi Mukherjee.

NCM Workshop on Statistical Learning

ISI Kolkata

Teaching Assistant

Jul 2024

National Centre for Mathematics (NCM) workshop for Ph.D. students and faculty.

Vectors & Matrices II

B.Stat 1st Year, ISI Kolkata

Co-Instructor

Jan–Jun 2024

Undergraduate course taught in collaboration with Prof. Sumitra Purakayastha.

Conferences & Workshops

ICTS-Networks Workshop: “Challenges in Networks”

ICTS-TIFR

Speaker & Research Group Member

Apr 2026

Working group (Week 1) and talk (Week 2) on joint research with Prof. Athreya and Prof. Bhattacharya.

Calcutta Triennial Symposium 2024

Kolkata, India

Participant

Dec 2024

Academic Service

Journal Referee: Served as a referee for *Sankhya B: The Indian Journal of Statistics* and the *Indian Journal of Pure and Applied Mathematics*.

Technical Skills

Languages: R, Python, C, $\text{\LaTeX}$

Tools: Overleaf, RStudio, MS Office

Extra-Curricular Activities

Music: Indian Classical Flute (under Sri Pankaj Banerjee) and Harmonica (under Sri Sukanta Ghoshal).